

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - SHORT ANSWER TEST

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5-Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 - Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse a real-world problem.		
Student Name:	DHIVYADHARSAN M.S	Reg. No.:	192511465
Section/Batch:	2025	Date:	04/09/2026

Code of Conduct

I DHIVYADHARSAN M.S (Name/RegNo)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

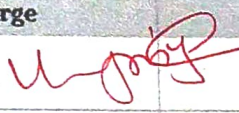
Signature of the Student: DHIVYADHARSAN M.S

Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answers using an analytical problem-solving, in evaluation the answers should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section/Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & Completion	10	Completes all questions accurately within the allotted time	Completes most questions on time	Requires additional time	Unable to complete the test

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature:	Signature:
Date:	Date:	Date:

SHORT ANSWER TEST

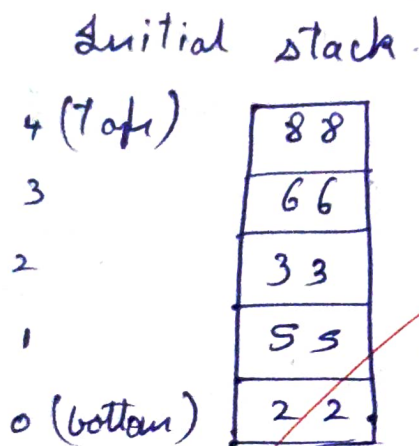
COS - AT3

Name: N.S. Dhivyadharsan
 Reg No: 192511485
 Course Code: CSA0305
 Course Name: Data Structure
 Date: 21/8/26

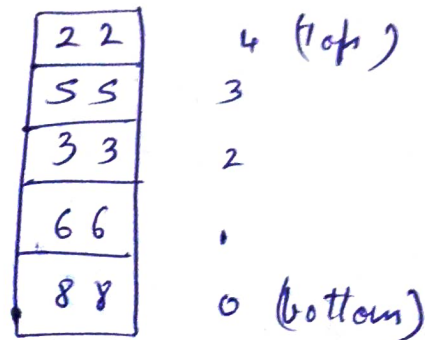
- 1) Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Construct ALL tree for the following elements.

Ans:

initial stack (size = 5)



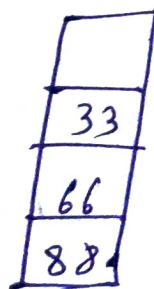
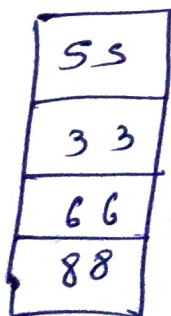
1) Invert The Elements



2) Pop() → 22

3) Pop() → 55

4) Pop() → 33



⇒ Remaining stack. bottom top: 88, 66

2) Linear Search Vs Binary Search

Ans: Elements 10, 20, 30, 40, 50, 60, 70, 80

Key = 70

⇒ Linear Search

10	20	30	40	50	60	70	80
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Comparisons

(10 → 20 → 30 → 40 → 50 → 60 → 70)

Time Complexity: $O(n)$

⇒ Binary Search mid

10	20	30	40	50	60	70	80
0	1	2	3	4	5	6	7

40 < 70

Go right

50	60	50	30	80
4	5	6	6	7

60 < 70

Go right

70	80
6	7

70 = 70

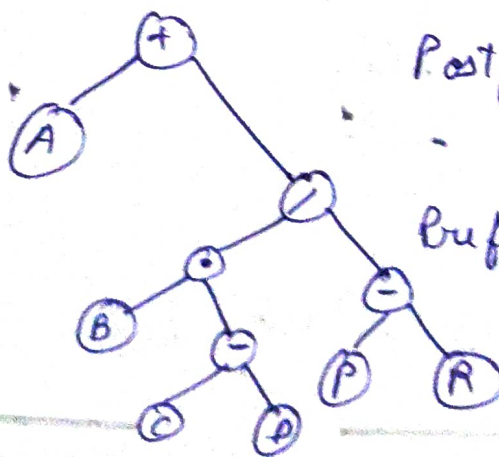
Found.

Comparisons: 3

Time Complexity: $O(\log n)$

3) Prefix And POST FIX

Ans: Expression $A + B * (C - D) / (P - R)$

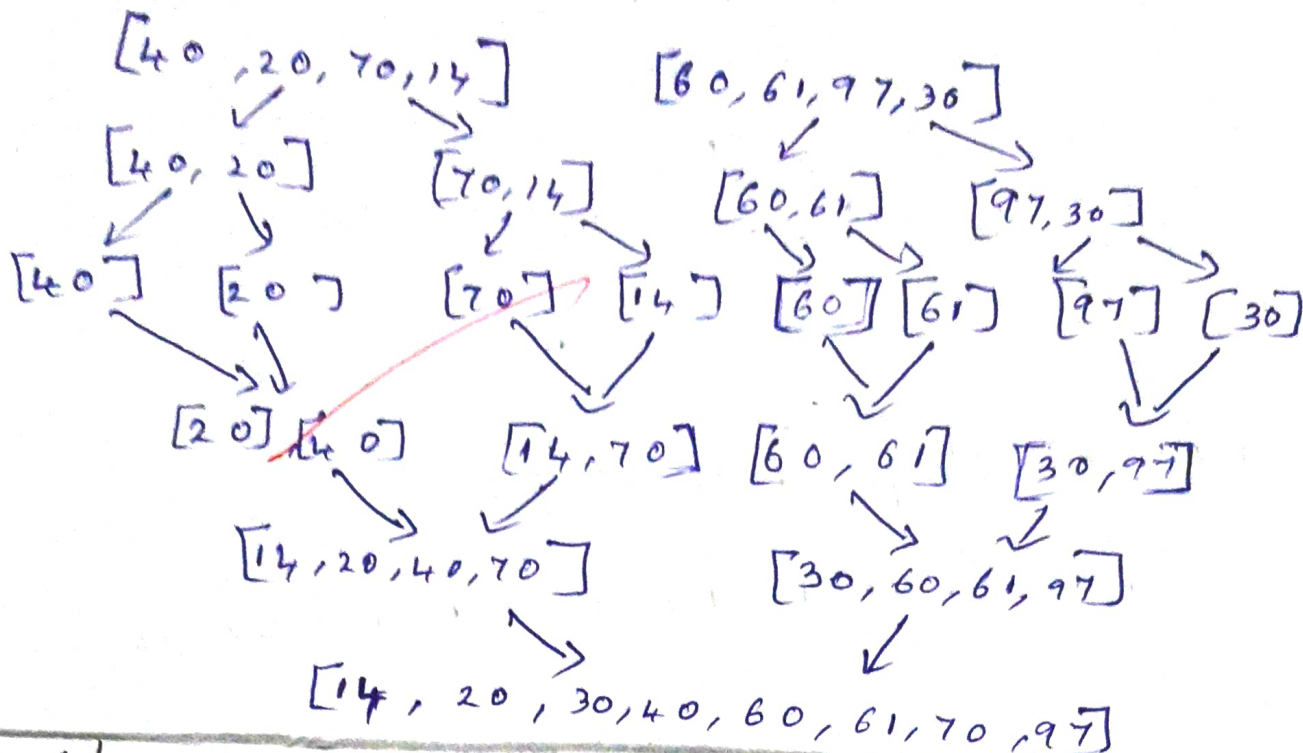


Postfix: $ABCD - * PR - / +$

Prefix: $+ A / * B - CD - PR$

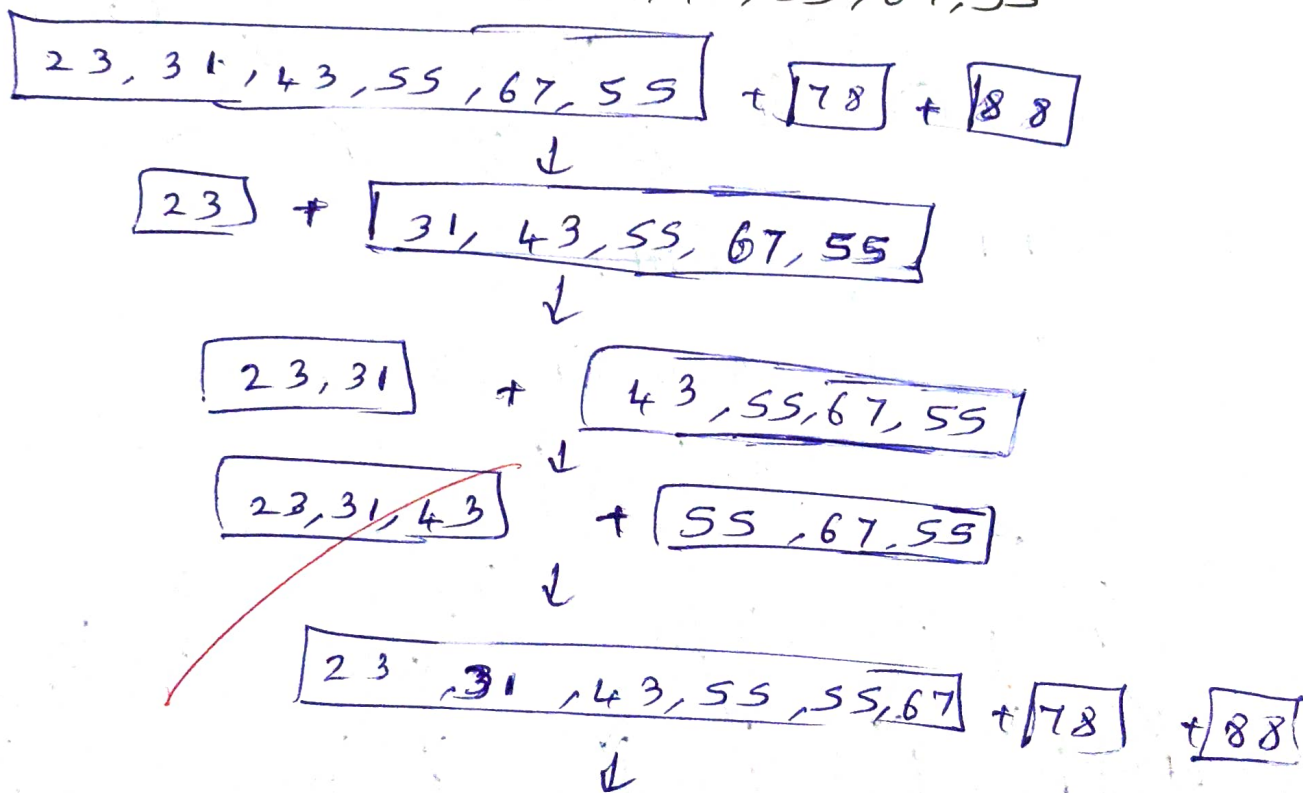
merge sort

Input : 40, 20, 70, 14, 60, 61, 77, 30



8) Quick sort

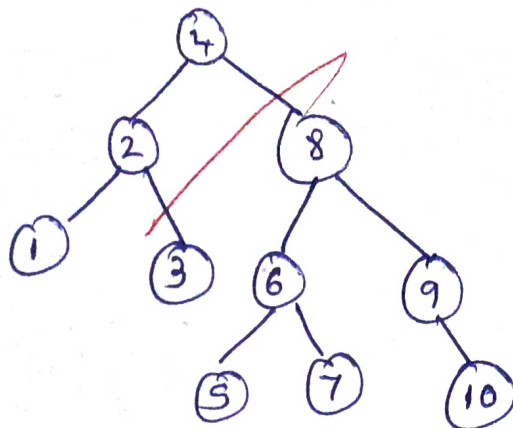
Input : 78, 23, 31, 88, 43, 55, 67, 55



Sorted result : 23, 31, 43, 55, 55, 67, 78, 88

7) AVL TREE (insert 1 to 10)

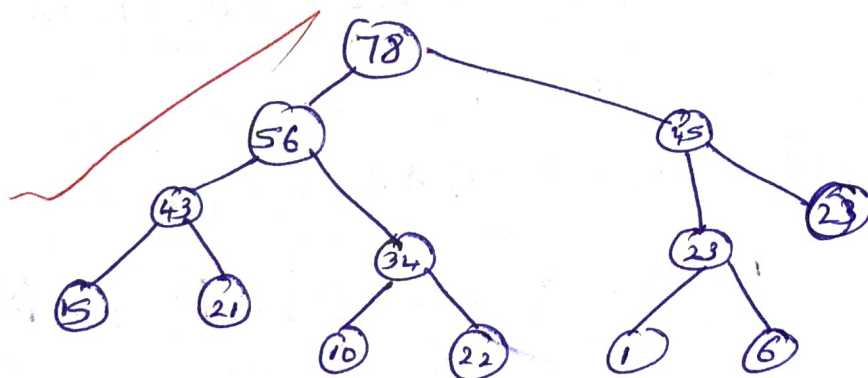
Ans:



5) Binary Heap (Max Heap)

Ans:

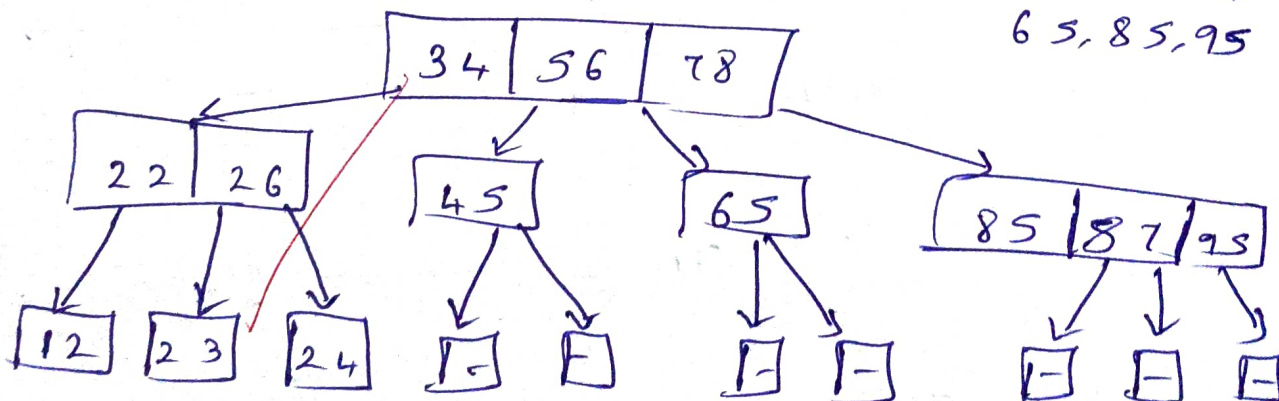
Insert: 10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45



6) B-TREE of ORDER 3

Ans:

Insert : 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95



(Each node has max 2 keys and max 3 children)

9) OPEN Hashing & Linear Probing

Ans: Given keys: 0, 1, 81, 4, 64, 25, 16, 36, 9, 49

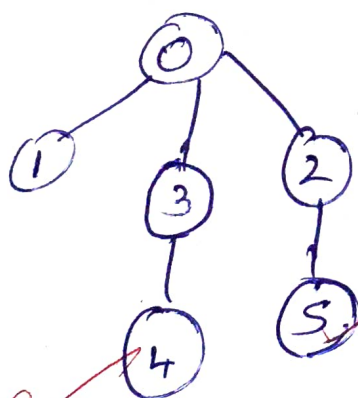
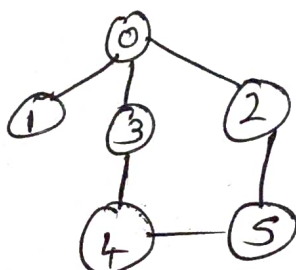
Hash function: $h(K) = K \bmod 10$

Hash Table (Size=10)

Index	0	1	2	3	4	5	6	7	8	9
Value	0	1	81	-	4	64	25	16	36	9*

→ 49 will be placed at next index 10 (if table size increased)

10) Consider the following graph & Construct BFS



Steps	Visited	Queue
1	0	1, 3, 2
2	1	3, 2
3	3	2, 4, 5
4	2	4, 5
5	4	5
6	5	-

BFS order: $0 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 5$